Johnbosco Ezekwe

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Open to relocation and remote work

Senior Software Engineer and Technical Lead with 10 years building national-scale health and fintech systems for underserved populations across Africa. I have led engineering teams of up to 13, including an 11-person regional team across 6 countries, and mentored 4 junior developers and 3 students. My platforms serve ministries of health and regulators in Botswana, Nigeria and Malawi, including a patient system live in 500+ facilities and a real-time HIV data pipeline covering 1.9M+ patients. I now bring the same secure, high-volume engineering to financial inclusion: invoice discounting for small suppliers and a mobile e-wallet.

Key Competencies: System Architecture • Distributed Systems • API Design & Development • Fintech & Payments • Health Informatics • Cloud Infrastructure • Data Systems • Engineering Leadership • Mentoring

Career Highlights:

- Leading a 9–13 person team developing Botswana's national Patient Information Management System (PIMS) for the Ministry of Health, live in 500+ health facilities.
- Re-architected data ingestion for Nigeria's national HIV repository (1.9M+ active patients) from a single-process batch job that took hours to days into a real-time, horizontally scalable RabbitMQ pipeline.
- Cut a production database bill by 75%, from approx. $4,000 to $1,000 a month, by migrating the AIMS datastore from Cosmos DB to MSSQL.
- Built NAFDAC's offline-first Mpox vaccine-safety platform, which enrolled approx. 4,000 participants (2× the 2,000 target) and submitted 1,853 reported safety cases to WHO VigiHub.
- Managed an 11-person regional engineering team across 6 African countries: Nigeria, Kenya, Botswana, Zambia, Malawi and Tanzania.

CORE TECHNOLOGIES

- **Languages:** C#, Python, Ruby, Java, JavaScript, VB.NET
- **Frameworks:** ASP.NET Core, Django, Ruby on Rails, React, React Native, Angular, Apache Cordova, Spring, Android (Java)
- **Databases:** MSSQL, PostgreSQL, MySQL, Redis, Cosmos DB, MongoDB
- **Messaging:** RabbitMQ
- **Cloud & DevOps:** AWS, Azure, Docker, Docker Swarm, CI/CD, Nginx
- **Architecture:** Microservices, queue-based distributed processing, multi-tenant SaaS, REST APIs, offline-first data collection
- **Security:** OAuth2 / OpenID Connect (IdentityServer4), JWT, RBAC
- **Testing:** xUnit (C#), Pytest, Rails tests, React component tests; unit and integration testing

- **Code quality:** Pylint, mypy, flake8, RuboCop, ESLint, EditorConfig, Git pre-commit and pre-push hooks

PROFESSIONAL EXPERIENCE

MGIC, University of Maryland Baltimore and BUMMHI are sister organizations in the University of Maryland global health network; I moved between them as programmes needed support.

Senior Software Engineer

EasyXchange Ltd – Gaborone, Botswana

May 2024 – Present · Part-time startup role, 20 hrs/week (evenings and weekends)

- One of three on the initial team (with the founder and one other engineer), owning architecture and delivery across all products.
- Built and launched exurl.co, a URL-shortening service in Python/Django with Redis caching for frequent redirects and PostgreSQL for link data. Containerized with Docker behind Nginx on AWS; runs at 99.9% uptime.
- Designed and built a multi-tenant invoice-discounting platform in Ruby on Rails that lets small suppliers get paid early on approved invoices, improving their access to working capital. Built authentication with role-based access, automated invoice workflows and financial reporting. Deployed using CI/CD on AWS; now onboarding client companies.
- Developing a microservices e-wallet and its React Native mobile app, with secure transactions, fraud monitoring and scalable payment processing, launching in 2027.
- Set up automated testing with Pytest and code-quality checks with RuboCop, Pylint, mypy, flake8 in the delivery pipeline.

Health Informatics / Software Development Lead

Botswana University of Maryland School of Medicine Health Initiative (BUMMHI) – Gaborone, Botswana

Apr 2023 – Present

- Took over Botswana's national Patient Information Management System (PIMS) from its original build team and lead a team of 9–13 engineers extending it, now live in 500+ health facilities. Work with the Ministry of Health, CDC Botswana, the Ministry of Local Government and health partners. Stack: VB.NET on MSSQL.
- Built a MongoDB-based middleware that connects PIMS to the Government of Botswana SMS gateway to send patient appointment reminders.
- Built a Patient Summary proof of concept with a React frontend and a C# ASP.NET backend on PostgreSQL.
- Designed StudySpace, a real-time monitoring platform used in Botswana's national Population-based HIV Impact Assessment (PHIA) survey to coordinate 300+ field teams and track survey progress. Stack: C# ASP.NET on MSSQL, deployed on AWS with Docker Swarm.
- Architected a national laboratory proficiency-testing platform used by 92 public and private laboratories across Botswana. Stack: C# ASP.NET on MSSQL.

Regional Software Development Manager

University of Maryland, Baltimore (UMB) – Remote, Africa region

Oct 2021 – Mar 2023

- Managed an 11-person engineering team across 6 countries, with at least one engineer per country (4 in Nigeria, 2–4 in Kenya).
- Led engineering for a COVID-19 vaccine cohort monitoring system in C# ASP.NET and MSSQL on Azure, tracking 25,000+ participants through 14+ follow-ups each (350,000+ scheduled visits). It scheduled follow-ups, sent clinicians their daily workload every morning, flagged missed visits and recorded reasons for loss to follow-up. Integrated with WHO's VigiFlow vaccine-safety platform.
- Designed national seroprevalence study systems covering 3,400+ households, with real-time data validation and reporting. Stack: C# ASP.NET and MSSQL on Azure.
- Architected AIMS (Activity Information Management System), a multi-country survey operations platform for PHIA HIV studies, working with CDC, national statistics offices and ministries of health in Nigeria, Zambia and Botswana. Offline-first field data collection, secured with OAuth2 (IdentityServer4). Cut its database bill by 75%, from ~$4,000 to ~$1,000 a month, by migrating from Cosmos DB to MSSQL.

Software Development Specialist

Maryland Global Initiatives Corporation (MGIC) – Abuja, Nigeria

Oct 2017 – Sep 2021

- Re-architected data ingestion for Nigeria's National Data Repository (NDR 3.0), covering 1.9M+ HIV patients, working with CDC/PEPFAR, the Federal Ministry of Health and implementing partners (APIN, CCFN, IHVN and others). Replaced a single-process job that took hours to days and often stalled with a real-time distributed pipeline: a RabbitMQ queue manager dispatches files to self-registering C# workers, each multithreaded with Parallel.ForEach and capped at 80% CPU. Capacity scales by adding workers.
- Built the XML extractor that sends patient records from NigeriaMRS (the national OpenMRS-based EMR, in Java/Spring) to the NDR, rolled out to facilities nationwide.
- Led Point of Care features for NigeriaMRS, integrating patient queue management and redesigning clinical forms; piloted in 5 facilities.
- Led a 3-person team building STaRT, a laboratory sample-tracking system piloted by Malawi's Ministry of Health. Stack: C# ASP.NET and MSSQL on Azure, with an Android (Java) mobile app.
- Built a logistics platform that coordinated laboratory supplies during the PHIA survey: a C# ASP.NET REST API on MSSQL with an Android (Java) mobile app.

Software Development Intern

MGIC – Abuja, Nigeria

Jan 2016 – Oct 2017

- Built an HIV Testing Services (HTS) Android app in Java for national testing data capture and reporting, backed by a C# REST API.
- Built MSERS, a disease surveillance and reporting system for the Nigeria Centre for Disease Control (NCDC), and AMC, a national drug-consumption reporting tool, each with a C# ASP.NET backend on MSSQL and an Android (Java) mobile app.
- Built Encuesta, a mobile survey app (Angular 2 on Apache Cordova) that generates data-collection forms dynamically from Excel templates, backed by a C# ASP.NET REST API on MSSQL.

CONSULTING & INDEPENDENT PROJECTS

Fixed-term remote contracts, delivered alongside full-time roles.

Data Manager – Mpox Vaccine CEM Monitoring System

National Agency for Food and Drug Administration and Control (NAFDAC) – Abuja, Nigeria (Remote)

Sep 2025 – Jun 2026 · Contract

- Designed and deployed an offline-first vaccine-safety monitoring platform: CSPro data collection in the field, syncing to ASP.NET dashboards. It enrolled approx. 4,000 participants and followed up 2,000.
- Built automated ingestion services that submitted 1,853 reported safety cases to the WHO Vigilance Hub.
- Deployed securely on NAFDAC's Azure infrastructure.

Consultant Engineer, CSPro

APIN Public Health Initiatives – Abuja, Nigeria (Remote)

Jan 2024 – Mar 2025 · Contract

- Designed and maintained CSPro data-collection apps for AIDS Impact Surveys.

ERP Software Consultant

Racida International – Nairobi, Kenya (Remote)

Dec 2021 – Apr 2022 · Contract

- Architected a full ERP system in C# ASP.NET covering HR, procurement, inventory, vendor management and performance tracking, holding records for 35 employees.

EDUCATION

- **MBA, Management Information Systems** – UNICAF University Zambia · Sep 2021
- **B.Eng, Mechanical Engineering** – Federal University of Technology Minna · Dec 2015